

Past Forward: Exif Revisited

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Abstract. Identifying the specific camera device used to capture a digital image is a cornerstone of image forensic investigation. This establishes a crucial link between a photograph and its source device, instrumental in verifying authenticity and determining provenance in legal and intelligence contexts. While analysis of Exchangeable Image File Format (Exif) metadata is efficient, these values are mutable and can be intentionally altered. This work presents a simple yet effective application capable of robustly identifying the original camera **Make** and **Model** of any photograph, regardless of metadata modification. To ensure integrity, we focus on analyzing the Exif structure of images from the Dresden and DivNoise camera forensics datasets, achieving promising results.

Keywords: Metadata · Image Verification · Exif Analysis.

1 Introduction

Camera identification, also known as device fingerprinting or device attribution, is recognized as a fundamental task in digital forensics [6]. Specifically, it involves reliably tracing an image back to its original source, particularly when the file has been intentionally modified. Digital image files contain Exif metadata, which is automatically embedded by the camera software and encompasses important details such as camera **Make**, camera **Model**, date/time, and other capture settings [9]. However, this type of metadata can be easily modified or stripped by third-party software for various purposes, including addressing privacy concerns (removing location tags), reducing file size, or malicious manipulation to conceal an image’s true origin. Since modification renders the simple metadata untrustworthy, there is a critical need for methods that look beyond the surface data to the underlying, intrinsic file characteristics. For more than two decades, research efforts have been made to address this problem, utilizing techniques such as sensor pattern noise [7], color filter array (CFA) interpolation [1], and deep learning [3]. However, no attempts have been made to directly classify camera devices based on the image Exif file structure and its detailed information. The ability to not only trace an image back to its original camera manufacturer

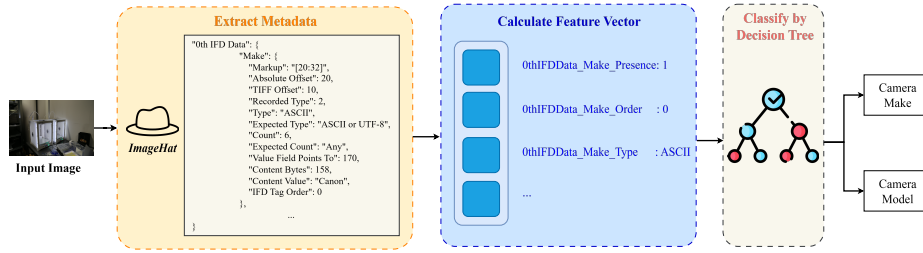


Fig. 1. The proposed method for camera identification.

and model but also to identify precisely which metadata tags have been altered significantly enhances the insights available for subsequent forensic tasks.

This paper introduces a simple yet effective system for camera identification, that leverages detailed image metadata features, specifically including the presence, order, and type of Exif fields. Specifically, the proposed approach capitalizes on the subtle differences in how various camera software handles metadata structuring. Furthermore, this work investigates the traces in features that remain when metadata is modified.

To facilitate feature extraction, our system utilizes a rich metadata reader to obtain the entire structure stored in the image Exif data. This means that images whose metadata have been stripped are not considered for camera classification. In contrast to many deep learning based methods, this approach offers transparency, allowing analysts to precisely trace the contribution of individual metadata features to the final classification prediction. In general, the contributions of this work are as follows:

- Implementation of a simple yet effective system using rich metadata to classify camera **Make** and camera **Model**.
- Evaluation of the proposed approach using two camera forensic datasets: Dresden [4] and DivNoise [2] datasets.
- Demonstration of a capability for detecting metadata tampering.

2 Proposed Method

2.1 Camera Identification

Our proposed approach (illustrated in Figure 1) utilizes only metadata information from the input image, specifically the presence, order, and type of fields that appear in the Exif file. Particularly, we deploy ImageHat⁵, a powerful Exif reader tool, to obtain the entire structure stored in image data, including file information, JPEG marker segments, Exif header structure, and individual IFD (Image File Directory) entries. Instead of relying on pixel-level characteristics such as sensor pattern noise or CFA traces, we extract metadata descriptors that reflect how a camera model encodes, organizes, and structures its metadata.

To construct feature representations, we parse the metadata and encode the following attributes:

⁵ <https://github.com/sinsaegit/ImageHat>

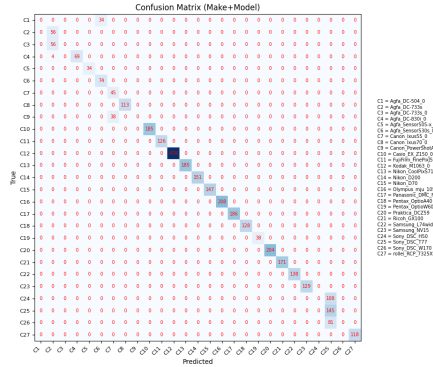


Fig. 2. Confusion matrix of **Make** and **Model** classifiers on Dresden dataset [4].

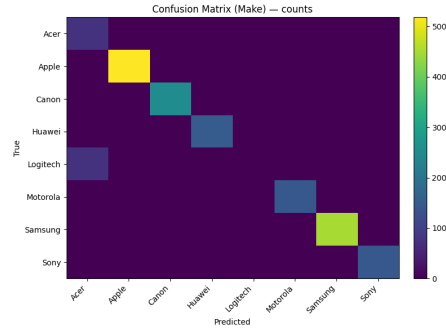


Fig. 3. Confusion matrix of camera **Make** classifiers on DivNoise dataset [2].

- The set of fields present in each dataset is recorded. Then, for every Exif file, each attribute in the block `BlockName_FieldName_Presence` (e.g., `Oth_IFD_Data_Make_Presence`, `EXIF_IFD_Data_Exposure_Time_Presence`), a value of 1 is assigned if the metadata contains the field inside the block; otherwise, it will be marked as 0.
- The attributes within `BlockName_FieldName_Order` receives ordering values within IFD tables.
- The attributes within `BlockName_FieldName_Type` receives integer values, which are enumerations of `SHORT`, `RATIONAL`, `ASCII`, etc.

These features are then transformed into fixed-length vectors, which are used to train standard classifiers (e.g., Decision tree [8], Random decision forest [5]). Importantly, since feature names are originally human-readable tags, the model decisions can be interpreted by tracing back predictions to specific entries. This allows forensic investigators to identify which tags or structures are most discriminative for a given camera model.

2.2 Metadata Editing Detection

Our camera identification method performs classification by analyzing the metadata structure, based on the finding that each camera manufacturer and software writes this data in a unique way. Likewise, metadata editing tools often read and rewrite fields differently from the original software, thereby altering the metadata structure. Hence, we demonstrate this difference by extracting and comparing metadata features in three scenarios: (1) original images versus images with edited metadata using PyExifTool⁶, (2) images before and after being uploaded to and re-downloaded from a social media platform, and (3) images before and after being processed with Adobe Photoshop⁷. In the latter two cases, the Exif data structure may be unintentionally modified.

⁶ <https://github.com/smarnach/pyexiftool>

⁷ https://www.adobe.com/vn_vi/products/photoshop/online.html

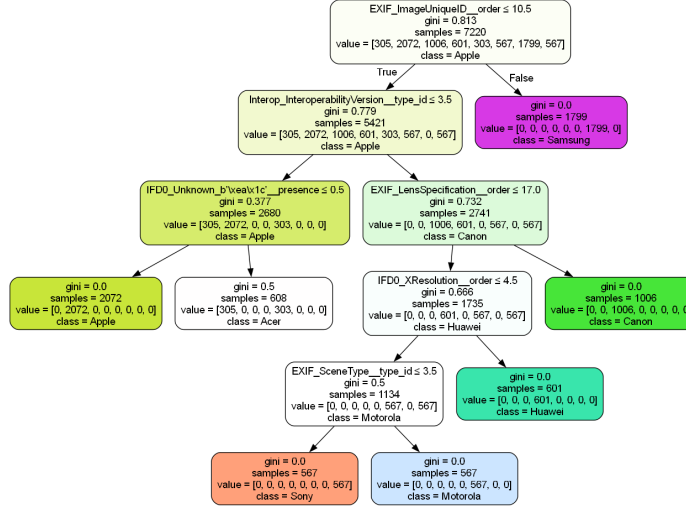


Fig. 4. Decision Tree classifier trained on DivNoise [2] dataset.

3 Experiments

3.1 Dataset

Dresden: This is an image database specifically built for the purpose of development and benchmarking of camera-based digital forensic techniques. The dataset comprises 17,242 images from 14 camera manufacturers: Agfa, Canon, Casio, Fujifilm, Kodak, Nikon, Olympus, Panasonic, Pentax, Praktica, Ricoh, Samsung, Sony, and Rollei. All images include complete metadata. For 5 camera makers (Agfa, Canon, Nikon, Pentax, Samsung), there are different camera models. There are 6 camera models under Agfa, 3 under Canon, 3 under Nikon, 2 under Pentax, and 2 under Samsung. Between the camera models, although they are from the same manufacturer, the metadata features are different.

DivNoise: This is a novel dataset designed to support the development of novel source identification techniques. It consists of image and video data from various device cameras, even data acquired from frontal cameras or webcams. This research utilizes 2,249 images from 8 different manufacturers: Acer, Apple, Canon, Huawei, Logitech, Motorola, Samsung, and Sony. For the 2 camera makers (Apple and Samsung), different metadata structures are found.

3.2 Results

Dresden: To evaluate model performance, the dataset was split into training and testing sets (80/20) using stratified sampling to preserve class distribution. Two classification models were trained and compared: Random Forest Classifier and Decision Tree Classifier. Both models achieved perfect classification performance, with all predictions matching the ground truth labels. This high accuracy

Table 1. Experimental Results for Metadata Editing Detection

Scenario Type	Case	Change
Social Media	Facebook	All metadata fields are stripped except Artist and Copyright
	X	All metadata fields are stripped
Metadata Editing Software	PyExifTool	No changes in structure
Photo Editing Software	Adobe Photoshop	Many changes in structure (both IFD blocks and fields)

is attributed to the rich and distinctive Exif metadata present in the dataset, which provides strong identifying features for each camera **Make**. The results confirm that camera **Make** can be reliably inferred using machine learning models trained solely on metadata. To classify up to the camera **Model**, we create labels by concatenating camera **Make** and camera **Model**, such as *Agfa_DC-504_0*, *Canon_PowerShotA640*. The classification results are visualized in Figure 2. As shown in the confusion matrix, some camera models are misclassified as they have the same metadata structure as others under the same manufacturer.

DivNoise: To evaluate model performance on the DivNoise dataset, the dataset was split into training and testing sets (80/20) using stratified sampling to maintain class distribution across camera makers. We trained a Random Forest Classifier and a Decision Tree Classifier, and they have the same classification results. Unlike the Dresden dataset, classification on DivNoise proved to be more challenging. While the model performed well for certain camera makers such as Apple, Canon, Huawei, Motorola, Samsung, and Sony, the overall accuracy was lower since Logitech was misclassified as Acer. The confusion matrix is shown in Figure 3, which suggests metadata from some devices is less distinctive. The decision tree classifier is visualized in Figure 4. The results of camera **Model** classification are the same as in Dresden. Typically, some camera models are misclassified as others under the same **Make** label. Exceptionally, *Logitech_C920_Frontal_0* is always misclassified as *Acer_AspireE5_Frontal_0*.

Metadata Editing Detection: We have discussed the need to demonstrate differences in metadata structure in various scenarios. Table 1 describes the changes before and after each scenario.

4 Demonstration

The demonstration aims to give users the experience of: (1) observing the decision process of camera **Make**, camera **Model** classification using metadata structure, (2) observing the differences/traces in metadata structure when edited or processed by PyExifTool, social media, and Adobe Photoshop. Figure 5 and figure 6 provide screen captures illustrating the system interface and workflow. Our demo video can be found at bit.ly/past_forward.

Camera Classification Demonstration

Classify camera make & model based on metadata structure

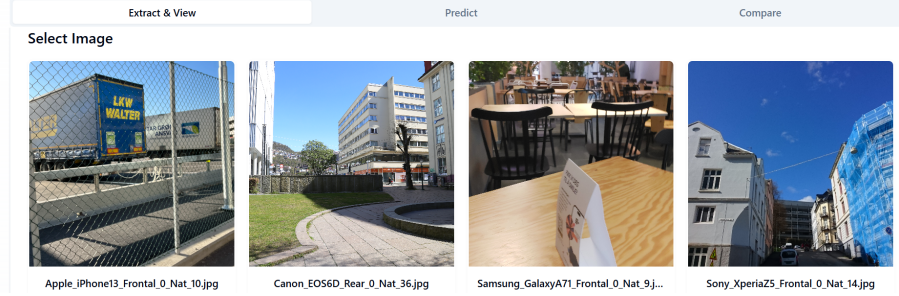


Fig. 5. Extract & View tab: select image and view metadata structure.

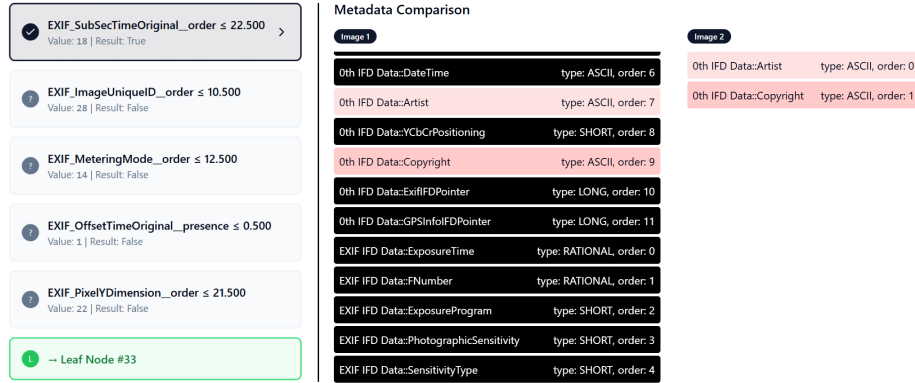


Fig. 6. Left side: Predict tab: visualize decision process of camera Make/Model prediction. Right side: Compare tab: compare metadata structures between two images.

5 Conclusion

Metadata structure can be used effectively and reliably for source attribution and integrity checks, such as consistency between structure and camera claims. Metadata edits and image post-processing can disrupt the initial structure, triggering a warning about metadata integrity. The current work, by exploring relatively narrow scenarios, suggests that each metadata structure and change can be attributed to specific cameras and software.

Acknowledgement We gratefully acknowledge the support of (1) the Intelligent Information Systems (I2S) research group, Department of Information Science and Media Studies at the University of Bergen, Norway, and (2) the European Union under Action 101158604 NORDIS, funded through the DIGITAL-2023-DEPLOY-04-EDMO-HUBS call by the European Health and Digital Executive Agency.

We would like to attribute MSc. Sindre Berg Sæter for supporting in the early stage of the project.

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